

SRS

# Software Requirements Specification

IEEE-830 style specification for Swiggy Nexus AI

|              |                                         |
|--------------|-----------------------------------------|
| Document     | SRS                                     |
| Version      | 1.0                                     |
| Status       | Approved                                |
| Owner        | Swiggy Nexus AI Team                    |
| Audience     | Engineering, Product, Design, Reviewers |
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*This document is part of the Swiggy Nexus AI reference implementation, an AI-powered personal commerce assistant built on the Swiggy MCP. It captures decisions, requirements, and design intent for engineers, reviewers, and stakeholders.*

# 1. Introduction

## 1.1 Purpose

This SRS specifies the functional and non-functional behavior of Swiggy Nexus AI v1.0. It is the authoritative contract between product, engineering, and QA.

## 1.2 Scope

The system provides a conversational interface that orchestrates Food, Instamart, and Dineout workflows through the Swiggy MCP, backed by persistent memory and analytics.

## 1.3 Definitions & Acronyms

| Term  | Meaning                                                          |
|-------|------------------------------------------------------------------|
| MCP   | Model Context Protocol — standardized tool interface for LLMs.   |
| Agent | LangGraph-based orchestrator that plans and executes tool calls. |
| SSE   | Server-Sent Events; used for streaming chat.                     |
| PKCE  | Proof Key for Code Exchange; OAuth 2.1 extension.                |

# 2. Overall Description

## 2.1 Product Perspective

Nexus AI sits between the end user and Swiggy's platform APIs. It does not replace Swiggy's checkout, catalog, or fulfillment — it wraps them behind a reasoning layer.

## 2.2 User Classes

- End Users — authenticated Swiggy customers.
- Developers — Builders Club members integrating the SDK.
- Administrators — internal operators with read-only audit access.

## 2.3 Assumptions & Dependencies

- Swiggy MCP is reachable and returns typed responses.
- OpenAI / Anthropic APIs are available with acceptable latency.
- Users have a valid Swiggy account.

# 3. Functional Requirements

| ID   | Requirement                                                             | Priority |
|------|-------------------------------------------------------------------------|----------|
| FR-1 | System shall authenticate users via OAuth 2.1 with PKCE.                | Must     |
| FR-2 | System shall accept natural language input and route to the agent.      | Must     |
| FR-3 | Agent shall build a cart draft and require confirmation before placing. | Must     |

| ID    | Requirement                                                                  | Priority |
|-------|------------------------------------------------------------------------------|----------|
| FR-4  | System shall stream tokens and tool calls via SSE.                           | Must     |
| FR-5  | System shall persist and retrieve long-term memory via pgvector.             | Must     |
| FR-6  | Users shall be able to view, edit, and delete stored memory.                 | Must     |
| FR-7  | System shall expose analytics dashboards (spend, nutrition, sustainability). | Should   |
| FR-8  | System shall support meal planning across a rolling 7-day window.            | Should   |
| FR-9  | System shall detect and warn on allergen conflicts.                          | Must     |
| FR-10 | System shall support price watch on Instamart items.                         | Could    |

4. Non-Functional Requirements

| Category        | Requirement                                                                            |
|-----------------|----------------------------------------------------------------------------------------|
| Performance     | p95 first-token < 1.2s; full plan < 4s; support 500 concurrent chat sessions per node. |
| Reliability     | 99.9% monthly uptime; RPO ≤ 5 min; RTO ≤ 30 min.                                       |
| Security        | OAuth 2.1 + PKCE; AES-256-GCM at rest; TLS 1.3 in transit; audited MCP calls.          |
| Scalability     | Horizontal autoscaling on CPU + custom chat_qps metric.                                |
| Usability       | WCAG 2.2 AA; keyboard navigable; ≥ 4.5:1 color contrast.                               |
| Maintainability | ≥ 95% coverage; strict typing; documented ADRs.                                        |
| Portability     | Runs in any CNCF-conformant K8s cluster; Docker Compose for local.                     |

5. External Interface Requirements

5.1 User Interfaces

Responsive web (mobile-first), streaming chat, cart preview, dashboard, analytics.

5.2 Software Interfaces

- Swiggy MCP — HTTPS, JSON-RPC 2.0.
- OpenAI / Anthropic — HTTPS REST + SSE.
- PostgreSQL 16 + pgvector — TCP.
- Redis 7 — TCP.

5.3 Communication Interfaces

- REST + SSE over HTTPS with HSTS.
- Webhooks for Swiggy order status updates.

## 6. Use Cases (Sample)

| Field         | Value                                                                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Use Case      | UC-01 Order Dinner via Chat                                                                                                             |
| Actor         | Authenticated End User                                                                                                                  |
| Precondition  | User is signed in and has a valid delivery address.                                                                                     |
| Main Flow     | 1. User sends intent. 2. Agent plans. 3. Agent calls MCP. 4. Cart preview shown. 5. User confirms. 6. Order placed. 7. Tracking begins. |
| Alt Flow      | 4a. User edits cart in chat; agent updates plan.                                                                                        |
| Postcondition | Order persisted; memory updated with episodic trace.                                                                                    |

## 7. Data Requirements

- All PII encrypted at rest with per-user envelope keys.
- Retention: chat 12 months, orders 84 months (compliance), memory user-controlled.
- Audit log is append-only and immutable.

## 8. Compliance

- DPDP Act, 2023 (India).
- PCI-DSS scope limited (payments handled by Swiggy).
- OWASP ASVS Level 2.